Valerii Kliuchnikov

Senior Software Engineer

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Profile

Senior Software Engineer (Go) with 9 years shipping production backend systems across fintech, AI tooling, and social platforms. Currently at Playneta building real-time LLM infrastructure at 9.6M messages/mo; previously delivered enterprise RBAC and payment systems at 2M+ daily transactions and 10K+ checks/sec with zero-downtime migrations for 1M+ users. Author of Engi, an open-source Go HTTP framework. Hands-on technical leadership: system design, 30+ technical interviews, 3 engineers mentored. Based in Barcelona, open to remote roles in EU and North America.

Technical Skills

Languages:	Go, SQL, Python, TypeScript
Backend:	Microservices, REST, gRPC, Protobuf, Gin, Echo, GORM, Wire DI, DDD, FastAPI, Strawberry
Data:	PostgreSQL, ScyllaDB, Cassandra, Redis
Messaging:	Kafka, Redpanda, RabbitMQ, asynq
AI/LLM:	MCP, OpenRouter, multi-model routing, Agno, Claude Code
DevOps:	GCP, Kubernetes, Helm, Docker, GitLab CI, GitHub Actions
Observability:	Prometheus, Grafana, OpenTelemetry
Security:	OAuth 2.0, JWT, RBAC
Compliance:	GDPR, PCI DSS

Professional Experience

May 2026 – Present — **Senior Software Engineer**, *B2B Contract, Playneta Limited*, Nicosia, Cyprus, Remote

- Owned system design and API development for a real-time LLM chat translation service in Go: ~10M messages/mo at sub-second median end-to-end; auto-detection across 26 source languages, targets selected dynamically from conversation roster; two-stage pipeline (Kafka consumer → asynq worker) hardened against hallucinations, truncation and concurrent updates
- Cost-optimised two-model LLM routing via OpenRouter (Gemini 2.5 Flash Lite primary, GPT-4.1 Nano fallback) with per-model circuit breaker and Redis cross-message cache, keeping total LLM spend under $600/month (~0.3% of platform revenue); full per-translation provenance (model, tokens, cost, latency) in Cassandra and Kafka events
- Translation infra reliability: LLM worker processes ~5M tasks/week at 99.99%+ success; HTTP API sustains 99.99%+ success at p99 read under 200ms; Helm-based Kubernetes with HPA across API, shared-consumer and worker pools
- Built a platform-wide UGC moderation service from scratch using strict DDD layering (Postgres + ScyllaDB + Redis, Wire DI); rule-based engine (Aho-Corasick dictionary + regex for contacts/links) covering nicknames, bio, club name/status and chat messages across 5 content types; bulk dictionary API with versioning, per-field rejection codes and a userId-scoped check endpoint
- Moderation serves ~100K checks/day at p99 under 15ms — well within the client fail-open deadline — with zero errors; integrated into user, chat and clubs so all text UGC and anti-spam routes through a single moderation path

- Designed and shipped a reusable DLQ layer for Redpanda consumers in the shared go-kiss2 library; adopted across 4 services (chat, clubs, translation, migration), eliminating silent message loss and capturing ∼2K failed messages/day into a single versioned dlq topic for triage and replay
- At-least-once semantics with per-group self-replay, configurable ReplayDelay/MaxReplays and a dedicated dlq.parked topic for exhausted messages; opt-in Permanent error marker lets handlers short-circuit non-retryable failures (schema violations, poison payloads) straight to parked, bypassing retry loops

August 2025 – April 2026 **Senior Software Engineer**, *B2B Contract, Refact.ai*, London, UK, Remote

- Built MCP server infrastructure on Kubernetes (isolated pod-per-integration model), declarative JSON preset system (0 → 30+ connectors: Slack, GitHub, Notion, Stripe, Supabase, Discord, etc.), and in-chat setup flow — reducing integration onboarding from manual k8s configuration to minutes
- Designed and shipped a complete OAuth 2.0 and token-based authorization framework covering 10+ external service providers (GitHub, Shopify, Google, and others)
- Authored `flexus-client-kit` — a Python SDK for agent developers with 15+ integration connectors (Gmail, Google Calendar/Sheets, Slack, GitHub, Jira, Telegram, Shopify, etc.) and abstractions for MCP tool registration, service lifecycle, and external auth flows

May 2024 – July 2025 **Senior Software Engineer**, *Ozon Bank*, Moscow, Russia, Remote

- Led system design and delivery of a Go-based RBAC platform (PostgreSQL, Redis, Kafka) serving 1M+ enterprise users under financial-regulator compliance; zero-downtime migration completed in 4 months, lifting permission-assignment success rate by 50%
- Built an automated data-sync pipeline in Go for 200K legal entities using event-driven architecture (Kafka + gRPC), cutting manual data updates by 60% and reducing staleness by 75%
- Architected a high-throughput notification service (Go, Kafka, PostgreSQL) processing 1M+ alerts/day at 99.9% uptime across multiple consumer groups
- Extended RBAC with real-time transaction-limit enforcement — 10K+ checks/sec over 1M active accounts — reducing limit-breach errors by 90%; designed for horizontal scaling on Kubernetes
- Conducted 30+ technical interviews, defined hiring rubrics, and mentored 3 engineers through structured onboarding, cutting ramp-up time by 40%

May 2022 – May 2024 **Software Engineer**, *Quadcode*, Larnaca, Cyprus, Hybrid

- Owned Go payment services layer (Redis, PostgreSQL, Kafka) processing up to 1M transactions/day across multiple crypto and fiat providers; optimised concurrency patterns cutting server load by 40% and reducing pipeline maintenance costs by 70%
- Prototyped an advanced Crypto Payment Provider integration, securing executive approval and rolling it out to three teams
- Designed a declarative integration framework that reduced new payment-provider onboarding from 2 weeks to 1 week
- Rebuilt GitLab CI with precommit hooks, accelerating pipelines by 20% and tightening code review cycles by 90%
- Delivered GDPR-compliant anonymization and PCI DSS tokenization services from scratch

Mar 2020 – May 2022 **Software Engineer**, *ProCSy*, Moscow, Russia, Remote

- Built 10+ Hyperledger Fabric smart contracts, automating workflows and reducing manual reconciliation by 70%
- Established an SMTP mailing system handling 10K+ daily emails at 98% delivery rate
- Decomposed a monolithic application into 6 microservices, cutting deployment time from 30 to 5 minutes
- Automated CI pipelines with GitLab CI and golangci-lint, achieving 90% reduction in feedback cycles

Jan 2018 – Mar 2020 **Junior Software Engineer**, *ITMO University*, Saint Petersburg, Russia, Hybrid

- Developed a modular resource-visualization system (Go, OOP) and a GORM-based relational schema for an internal simulation platform used in research courseware

Projects

Engi Lightweight Go HTTP framework: declarative routing, structured services, automatic parameter validation, auth and CORS middleware, OpenTelemetry tracing — and auto-generated OpenAPI documentation derived from route declarations. Open-source; GoDoc, CI, Go Report Card.
`github.com/kliuchnikovv/engi`

Nomi End-to-end personal AI agent system: Go/Gin API gateway, Python/Agno agent orchestration (tool use, multi-agent teams), React Native mobile client. Plugin architecture — Gmail, Calendar, GitHub, Slack connectors. Deployed on Railway with Docker; 1,700+ commits.
`github.com/kliuchnikovv/nomi`

Keystone Local-first smart-home hub on ESP32 (C++/ESP-IDF): multi-chip architecture — brain (S3) runs the entity registry, state machine and rules engine; radio co-processors handle Zigbee (C6), Matter and BLE. Go relay server bridges hub to internet. React Native mobile app over authenticated WebSocket/JSON protocol. OTA firmware updates.
`github.com/kliuchnikovv/keystone`

DnD Detective RPG with a deterministic Go rules engine and an LLM narrator (OpenRouter/Anthropic). Trust-boundary architecture: model only narrates what the core decided — zero hallucinated facts. HTTP/WebSocket server, React Native mobile client (iOS-first), pluggable rule systems (D&D 5e, Cyberpunk RED). 188K lines Go, 1,211 tests, 10 ADRs.
`github.com/kliuchnikovv/dnd`

Education

Sep 2015 – Jun 2019 **Bachelor of Science in Software Engineering**
Faculty of Software Engineering and Computer Systems, *ITMO University*, Saint Petersburg, Russia

Additional Information

Languages Russian — Native
English — Professional working proficiency

Interests 3D modelling & printing, RC aircraft & cars, tabletop RPG, smart home, gaming